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METHOD FOR PROVIDING SERVICES WITH MULTIPLE VISUAL AGENTS WITH ARTIFICIAL INTELLIGENCE

Abstract

Embodiments of the present disclosure may include a method for providing services via a set of virtual agents with artificial intelligence in a service area.

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Background/Summary

BACKGROUND OF THE INVENTION

[0001] Embodiments of the present disclosure may include a method for providing services via multiple virtual agents with artificial intelligence in a service area.

BRIEF SUMMARY

[0002] Embodiments of the present disclosure may include a method for providing sales or services via a set of virtual agents with artificial intelligence in a service area, the method including detecting, by one or more processors, a parking request from a user with the agents with artificial intelligence. In some embodiments, the user may drive a vehicle into a garage of a mall. Embodiments may also include activating the first visual agent of the set of visual agents with artificial intelligence.

[0003] In some embodiments, the first visual agent may be configured to be a garage security personnel. In some embodiments, the first visual agent may be configured to greet the user, give a parking spot, and record the location of the parking spot. In some embodiments, the first visual agent may be configured to give directions of a first place that the user wants to visit.

[0004] In some embodiments, the set of visual agents may be empowered by an artificial intelligence engine. In some embodiments, the set of visual agents may be configured to collaborate. In some embodiments, an artificial intelligence engine may be coupled to the one or more processors and a server. In some embodiments, the artificial intelligence engine may be trained by human experts in the field.

[0005] In some embodiments, the set of virtual agents may be configured to be displayed in LED/OLED displays, Android/iOS tablets, Laptops/PCs, smartphones, or VR/AR goggles. In some embodiments, a set of multi-layer info panels coupled to the one or more processors may be configured to overlay graphics on top of the set of virtual agents. In some embodiments, any of the set of virtual agents may be configured to be displayed with an appearance of a real human or a humanoid or a cartoon character.

[0006] In some embodiments, any of the set of virtual agent's gender, age, and ethnicity may be determined by the artificial Intelligence's analysis of input from the user. In some embodiments, any of the set of virtual agents may be configured to be displayed in full-body or half-body portrait mode. In some embodiments, the artificial intelligence engine may be configured for real-time speech recognition, speech-to-text generation, real-time dialog generation, text-to-speech generation, voice-driven animation, and human avatar generation.

[0007] In some embodiments, the artificial intelligence engine may be configured to emulate different voices and use different languages. Embodiments may also include activating, at the first place, by one or more processors, a second visual agent of the set of visual agents, when the user arrives via the directions of the first place given by the first visual agent.

[0008] In some embodiments, the second visual agent may be configured to communicate with the first visual agent on the info of the user. In some embodiments, the first place may be a movie theater. In some embodiments, the second visual agent may be configured to greet and display the movie time slots and seat availability in the theater.

[0009] In some embodiments, the second visual agent may be configured to recommend movies based on the user's past choices. In some embodiments, the second visual agent may be configured to pick a time and a seat for the user for the movies based on the user's choice. In some embodiments, the second visual agent may be configured to give directions to a second place that the user wants to visit.

[0010] In some embodiments, the second visual agent may be configured to offer validation of parking tickets. Embodiments may also include activating, at the second place, by one or more processors, a third visual agent of the set of visual agents, when the user arrives via the directions of the second place given by the second visual agent. In some embodiments, the third visual agent may be configured to communicate with the first visual agent and the second visual agent on info of the user.

[0011] In some embodiments, the second place may be a restaurant. In some embodiments, the third visual agent may be configured to greet and recommend movies based on the user's past choices. In some embodiments, the third visual agent may be configured to complete the food

transaction with the user. In some embodiments, the third visual agent may be configured to give directions of destination chosen by the user.

[0012] Embodiments may also include activating the first visual agent of the set of visual agents with artificial intelligence, at the garage of the mall, when the user arrives at the garage. In some embodiments, the first visual agent may be configured to recognize the user and the validated parking ticket. In some embodiments, the first visual agent may be configured to say goodbye and welcome to visit us again in the future to the user when the user drives away.

[0013] Embodiments of the present disclosure may also include a method for providing services via a set of virtual agents with artificial intelligence in a service area, the method including activating a set of visual agents at different locations with artificial intelligence sequentially. In some embodiments, any of the visual agents may be configured to greet the user.

[0014] In some embodiments, the set of visual agents may be empowered by an artificial intelligence engine. In some embodiments, the set of visual agents may be configured to collaborate with each other. In some embodiments, an artificial intelligence engine may be coupled to the one or more processors and a server. In some embodiments, the artificial intelligence engine may be trained by human experts in the field.

[0015] In some embodiments, the set of virtual agents may be configured to be displayed in LED/OLED displays, Android/iOS tablets, Laptops/PCs, smartphones, or VR/AR goggles. In some embodiments, a set of multi-layer info panels coupled to the one or more processors may be configured to overlay graphics on top of the set of virtual agents. In some embodiments, any of the set of virtual agents may be configured to be displayed with the appearance of a real human, a humanoid or a cartoon character.

[0016] In some embodiments, any of the set of virtual agent's gender, age, and ethnicity may be determined by the artificial Intelligence's analysis of input from the user. In some embodiments, any of the set of virtual agents may be configured to be displayed in full-body or half-body portrait mode. In some embodiments, the artificial intelligence engine may be configured for real-time speech recognition, speech-to-text generation, real-time dialog generation, text-to-speech generation, voice-driven animation, and human avatar generation.

[0017] In some embodiments, the artificial intelligence engine may be configured to emulate different voices and use different languages. Embodiments may also include enabling the set of visual agents at different locations with artificial intelligence sequentially to assist the user with a set of different requests. In some embodiments, the individual agent of the set of visual agents may be configured to be optimized with tasks related to business that the individual agent may be located.

[0018] Embodiments may also include enabling the set of visual agents at different locations with artificial intelligence to exchange and analyze the information of the user to better assist the user. Embodiments may also include recording interactions between the user and the set of visual agents. Embodiments may also include analyzing the interactions between the user and the set of visual agents to better assist the user in the next encounter with the user.

[0019] Embodiments of the present disclosure may also include a system for providing services via a set of virtual agents with artificial intelligence in a service area, the system including a set of visual agents at different locations with artificial intelligence sequentially. In some embodiments, any of the visual agents may be configured to greet the user.

[0020] In some embodiments, the set of visual agents may be configured to collaborate among the set of visual agents. In some embodiments, the different locations may be physical locations or digital locations. Embodiments may also include an artificial intelligence engine that may be coupled to the one or more processors a server and a set of visual agents.

[0021] In some embodiments, the set of visual agents may be empowered by the artificial intelligence engine. In some embodiments, the artificial intelligence engine may be trained by human experts in the field. In some embodiments, the set of virtual agents may be configured to be

displayed in LED/OLED displays, Android/iOS tablets, Laptops/PCs, smartphones, or VR/AR goggles.

[0022] In some embodiments, a set of multi-layer info panels coupled to one or more processors may be configured to overlay graphics on top of the set of virtual agents. In some embodiments, any of the set of virtual agents may be configured to be displayed with the appearance of a real human, a humanoid or a cartoon character. In some embodiments, any of the set of virtual agent's gender, age, and ethnicity may be determined by the artificial Intelligence's analysis of input from the user.

[0023] In some embodiments, any of the set of virtual agents may be configured to be displayed in full-body or half-body portrait mode. In some embodiments, the artificial intelligence engine may be configured for real-time speech recognition, speech-to-text generation, real-time dialog generation, text-to-speech generation, voice-driven animation, and human avatar generation.

[0024] In some embodiments, the artificial intelligence engine may be configured to emulate different voices and use different languages. In some embodiments, the artificial intelligence engine may be configured to perform the following tasks enabling the set of visual agents at different locations with artificial intelligence sequentially to assist the user with a set of different requests.

[0025] In some embodiments, the individual agent of the set of visual agents may be configured to be optimized with tasks related to the business that the individual agent may be located. Embodiments may also include enabling the set of visual agents at different locations with artificial intelligence to exchange and analyze the information of the user to better assist the user. Embodiments may also include recording interactions between the user and the set of visual agents. Embodiments may also include analyzing the interactions between the user and the set of visual agents to better assist the user in the next encounter with the user.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0026] FIG. 1 is a flowchart illustrating a method for providing services via multiple virtual agents with artificial intelligence, according to some embodiments of the present disclosure.

[0027] FIG. 2 is a flowchart illustrating a method for providing services via multiple virtual agents with artificial intelligence, according to some embodiments of the present disclosure.

[0028] FIG. 3 is a block diagram illustrating a system, according to some embodiments of the present disclosure.

[0029] FIG. 4 is a diagram showing an example of a method for providing services via multiple virtual agents with artificial intelligence, according to some embodiments of the present disclosure.

[0030] FIG. 5 is a diagram showing a second example of a method for providing embodiments of the present disclosure.

[0031] FIG. 6 is a diagram showing a third example of a method for providing services via multiple virtual agents with artificial intelligence, according to some embodiments of the present disclosure.

[0032] FIG. 7 is a diagram showing a fourth example of a method for providing services via multiple virtual agents with artificial intelligence, according to some embodiments of the present disclosure.

[0033] FIG. 8 is a diagram showing a fifth example of a method for providing services via multiple virtual agents with artificial intelligence, according to some embodiments of the present disclosure.

[0034] FIG. 9 is a diagram showing a sixth example of a method for providing services via multiple virtual agents with artificial intelligence, according to some embodiments of the present disclosure.

[0035] FIG. 10 is a diagram showing a seventh example of a method for providing services via

multiple virtual agents with artificial intelligence, according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0036] FIG. 1 is a flowchart that describes a method for providing services via multiple virtual agents with artificial intelligence, according to some embodiments of the present disclosure. In some embodiments, at **110**, the method may include detecting, by one or more processors, a parking request from a user. At **120**, the method may include activating a first visual agent of the set of visual agents with artificial intelligence. At **130**, the method may include activating, at the first place, by one or more processors, a second visual agent of the set of visual agents, when the user arrives via the directions of the first place given by the first visual agent.

[0037] In some embodiments, at **140**, the method may include activating, at the second place, by one or more processors, a third visual agent of the set of visual agents, when the user arrives via the directions of the second place given by the second visual agent. At **150**, the method may include activating the first visual agent of the set of visual agents with artificial intelligence, at the garage of the mall, when the user arrives at the garage.

[0038] In some embodiments, the user may drive a vehicle into a garage of a mall. The first visual agent may be configured to be a garage security personnel. The first visual agent may be configured to greet the user, give a parking spot and record location of the parking spot. The first visual agent may be configured to give directions of a first place that the user may want to visit. The set of visual agents may be empowered by an artificial intelligence engine.

[0039] In some embodiments, the set of visual agents may be configured to collaborate with each other. An artificial intelligence engine may be coupled to the one or more processors and a server. The artificial intelligence engine may be trained by human experts in the field. The set of virtual agents may be configured to be displayed in LED/OLED displays, Android/iOS tablets, Laptops/PCs, smartphones, or VR/AR goggles. A set of multi-layer info panels coupled to the one or more processors may be configured to overlay graphics on top of the set of virtual agents.

[0040] In some embodiments, any of the set of virtual agents may be configured to be displayed with the appearance of a real human, a humanoid or a cartoon character. Any of the set of virtual agent's gender, age, and ethnicity may be determined by the artificial Intelligence's analysis of input from the user. Any of the set of virtual agents may be configured to be displayed in full-body or half-body portrait mode. The artificial intelligence engine may be configured for real-time speech recognition, speech-to-text generation, real-time dialog generation, text-to-speech generation, voice-driven animation, and human avatar generation.

[0041] In some embodiments, the artificial intelligence engine may be configured to emulate different voices and use different languages. The second visual agent may be configured to communicate with the first visual agent on the info of the user. The first place may be a movie theater. The second visual agent may be configured to greet and display the movie time slots and seat availability in the theater. The second visual agent may be configured to recommend movies based on the user's past choices.

[0042] In some embodiments, the second visual agent may be configured to pick a time and a seat for the user for the movies based on user's choice. The second visual agent may be configured to give directions to a second place that the user may want to visit. The second visual agent may be configured to offer validation of parking ticket. The third visual agent may be configured to be communicating with the first visual agent and the second visual agent on info of the user.

[0043] In some embodiments, the second place may be a restaurant. The third visual agent may be configured to greet and recommend movies based on the user's past choices. The third visual agent may be configured to complete the food transaction with the user. The third visual agent may be configured to give directions of destination chosen by the user. The first visual agent may be configured to recognize the user and the validated parking ticket. The first visual agent may be configured to say goodbye and welcome to visit us again in the future to the user when the user

may drive away.

[0044] FIG. 2 is a flowchart that describes a method for providing services via multiple virtual agents with artificial intelligence, according to some embodiments of the present disclosure. In some embodiments, at **210**, the method may include activating a set of visual agents at different locations with artificial intelligence sequentially. At **220**, the method may include enabling the set of visual agents at different locations with artificial intelligence sequentially to assist the user with a set of different requests. At **230**, the method may include enabling the set of visual agents at different locations with artificial intelligence to exchange and analyze the information of the user to better assist the user. At **240**, the method may include recording interactions between the user and the set of visual agents. At **250**, the method may include analyzing the interactions between the user and the set of visual agents to better assist the user in next encounter with the user.

[0045] In some embodiments, any of the visual agents may be configured to greet the user. The set of visual agents may be empowered by an artificial intelligence engine. The set of visual agents may be configured to collaborate with each other. An artificial intelligence engine may be coupled to the one or more processors and a server. The artificial intelligence engine may be trained by human experts in the field. The set of virtual agents may be configured to be displayed in LED/OLED displays, Android/iOS tablets, Laptops/PCS, smartphones, or VR/AR goggles.

[0046] In some embodiments, a set of multi-layer info panels coupled to one or more processors may be configured to overlay graphics on top of the set of virtual agents. Any of the set of virtual agents may be configured to be displayed with the appearance of a real human, a humanoid or a cartoon character. Any of the set of virtual agent's gender, age and ethnicity may be determined by the artificial Intelligence's analysis of input from the user.

[0047] In some embodiments, any of the set of virtual agents may be configured to be displayed in full-body or half-body portrait mode. The artificial intelligence engine may be configured for real-time speech recognition, speech-to-text generation, real-time dialog generation, text-to-speech generation, voice-driven animation, and human avatar generation. The artificial intelligence engine may be configured to emulate different voices and use different languages. The individual agent of the set of visual agents may be configured to be optimized with tasks related to the business that the individual agent may be located.

[0048] FIG. 3 is a block diagram that describes a system **300**, according to some embodiments of the present disclosure. In some embodiments, the system **300** may include a set of visual agents **310** at different locations with artificial intelligence sequentially, an artificial intelligence engine **320** that may be coupled to the one or more processors and a server and the set of visual agents **310**, and recording interactions **330** between the user and the set of visual agents **310**. Any of the visual agents **310** may be configured to greet the user.

[0049] In some embodiments, the set of visual agents **310** may be configured to collaborate among the set of visual agents **310**. The different locations may be physical locations or digital locations. The set of visual agents **310** may be empowered by the artificial intelligence engine **320**. The artificial intelligence engine **320** may be trained by human experts in the field. The set of virtual agents may be configured to be displayed in LED/OLED displays, Android/iOS tablets, Laptops/PCs, smartphones, or VR/AR goggles.

[0050] In some embodiments, a set of multi-layer info panels coupled to the one or more processors may be configured to overlay graphics on top of the set of virtual agents. Any of the set of virtual agents may be configured to be displayed with the appearance of a real human or a humanoid or a cartoon character. Any of the set of virtual agent's gender, age and ethnicity may be determined by the artificial Intelligence's analysis on input from the user.

[0051] In some embodiments, any of the set of virtual agents may be configured to be displayed in full body or half body portrait mode. The artificial intelligence engine **320** may be configured for real-time speech recognition, speech-to-text generation, real-time dialog generation, text-to-speech generation, voice-driven animation, and human avatar generation. The artificial intelligence engine

320 may be configured to emulate different voices and use different languages.

[0052] In some embodiments, the artificial intelligence engine **320** may be configured to perform the following tasks: Enabling the set of visual agents **310** at different locations with artificial intelligence sequentially to assist the user with a set of different requests. The individual agent of the set of visual agents **310** may be configured to be optimized with tasks related to the business where the individual agent may be located. Enabling the set of visual agents **310** at different locations with artificial intelligence to exchange and analyze the information of the user to better assist the user. Analyzing the interactions **330** between the user and the set of visual agents **310** to better assist the user in the next encounter with the user.

[0053] FIG. **4** is a diagram showing an example that describes a first example of a method for providing services via multiple virtual agents with artificial intelligence, according to some embodiments of the present disclosure.

[0054] In some embodiments, a user **405** can approach a smart display **410**. In some embodiments, the smart display **410** could be LED or OLED-based. In some embodiments, interactive panels **420** are attached to the smart display **410**. In some embodiments, camera **425**, sensor **430** and microphone **435** are attached to the smart display **410**. In some embodiments, an artificial intelligence visual assistant **415** is active on the smart display **410**. In some embodiments, a visual working agenda **460** is shown on the smart display **410**. In some embodiments, user **405** can approach the smart display **410** and initiate and complete the intended business with the visual assistant **415** by the methods described in FIG. **1**-FIG. **3**. In some embodiments, interactive panel **420** is coupled to a central processor. In some embodiments, interactive panel **420** is coupled to a server via a wireless link. In some embodiments, user **405** can interact with the visual assistant **415** via camera **425**, sensor **430** and microphone **435** using methods described in FIG. **1**-FIG. **3**, with the help of interactive panel **420**. In some embodiments, user **405** can choose what language to use. In some embodiments, other users can use this service described in this paragraph. In some embodiments, the user is able to interact with multiple AI visual agents as described in this example and methods described in FIG. **1**-**3**.

[0055] FIG. **5** is a diagram showing a second example of a method for providing embodiments of the present disclosure.

[0056] In some embodiments, a user **505** can approach a smart display **510**. In some embodiments, the smart display **510** could be LED or OLED-based. In some embodiments, interactive panels **520** are attached to the smart display **510**. In some embodiments, camera **525**, sensor **530**, and microphone **535** are attached to the smart display **510**. In some embodiments, a support column **550** is attached to the smart display **510**. In some embodiments, an artificial intelligence visual assistant **515** is active on the smart display **510**. In some embodiments, a visual working agenda **560** is shown on the smart display **510**. In some embodiments, user **505** can approach the smart display **510** and initiate and complete the business process with the visual assistant **515** by the methods described in FIG. **1**-FIG. **3**. In some embodiments, interactive panel **520** is coupled to a central processor. In some embodiments, interactive panel **520** is coupled to a server via a wireless link. In some embodiments, user **505** can interact with the visual assistant **515** via camera **525**, sensor **530** and microphone **535** using methods described in FIG. **1**-FIG. **3**, with the help of interactive panel **520**. In some embodiments, user **505** can choose what language to be used. In some embodiments, other users can use this service described in this paragraph. In some embodiments, other users can use this service described in this paragraph. In some embodiments, the user is able to interact with multiple AI visual assistants as described in this example and methods described in FIG. **1**-**3**.

[0057] FIG. **6** is a diagram showing a third example of a method for providing services via multiple virtual agents with artificial intelligence, according to some embodiments of the present disclosure.

[0058] In some embodiments, a user **605** can approach a smart display **610**. In some embodiments, the smart display **610** could be LED or OLED-based. In some embodiments, the display **610** could

be a part of a desktop computer, a laptop computer or a tablet computer. In some embodiments, a camera, sensor, and microphone are attached to the smart display **610**. In some embodiments, an artificial intelligence visual assistant **615** is active on the smart display **610**. In some embodiments, a visual working agenda **660** is shown on the smart display **610**. In some embodiments, user **605** can approach the smart display **610** and initiate and complete the business process with the visual assistant **615** by the methods described in FIG. 1-FIG. 3. In some embodiments, a keyboard is coupled to a central processor. In some embodiments, a keyboard is coupled to a server via a wireless link. In some embodiments, user **605** can interact with the visual assistant **615** via a camera, sensor and microphone using methods described in FIG. 1-FIG. 3, with the help of the keyboard. In some embodiments, user **605** can choose what language to use. In some embodiments, other users can use this service described in this paragraph. In some embodiments, other users can use this service described in this paragraph. In some embodiments, the user is able to interact with multiple AI visual assistants as described in this example and methods described in FIG. 1-3.

[0059] FIG. 7 is a diagram showing a fourth example of a method for providing services via multiple virtual agents with artificial intelligence, according to some embodiments of the present disclosure.

[0060] In some embodiments, a user **705** can view programs including news with a VR or AR device **710**. In some embodiments, a processor and a server are connected to the VR or AR device **710**. In some embodiments, an interactive keyboard is connected to the VR or AR device **710**. In some embodiments, an AI visual assistant **715** is active on the VR or AR device **710**. In some embodiments, a visual working agenda **760** is shown on the VR or AR **710**. In some embodiments, user **705** can initiate and complete the business process with the visual assistant **705** via the VR or AR device **715** by the methods described in FIG. 1-FIG. 3. In some embodiments, an interactive panel is coupled to a central processor. In some embodiments, an interactive panel is coupled to a server via a wireless link. In some embodiments, the user **705** can choose what language to use. In some embodiments, other users can use this service described in this paragraph. In some embodiments, other users can use this service described in this paragraph. In some embodiments, the user is able to interact with multiple AI visual assistants as described in this example and methods described in FIG. 1-3.

[0061] FIG. 8 is a diagram showing a fifth example of a method for providing services via multiple virtual agents with artificial intelligence, according to some embodiments of the present disclosure.

[0062] In some embodiments, a user **805** can view programs including news with a smartphone device **810**. In some embodiments, a processor and a server are connected to the smartphone device **810**. In some embodiments, an interactive keyboard is connected to the smartphone device **810**. In some embodiments, an AI visual assistant **815** is active on the smartphone device **810**. In some embodiments, a visual working agenda **860** is shown on the smartphone device **810**. In some embodiments, user **805** can initiate and complete the business process with the visual assistant **815** via smartphone device **810** by the methods described in FIG. 1-FIG. 3. In some embodiments, an interactive panel is coupled to a central processor. In some embodiments, interactive panel is coupled to a server via a wireless link. In some embodiments, the user **805** can choose what language to be used. In some embodiments, other users can use this service described in this paragraph. In some embodiments, other users can use this service described in this paragraph. In some embodiments, the user is able to interact with multiple AI visual assistants as described in this example and methods described in FIG. 1-3.

[0063] FIG. 9 is a diagram showing a sixth example of a method for providing services via multiple virtual agents with artificial intelligence, according to some embodiments of the present disclosure.

[0064] In some embodiments, a user **905** has a brain-computer interface. In some embodiments, the user **905** may wear a headset **907** that can detect and translate the electric signal from the brain and communicate with the computer or other devices. The computer **910** or other devices are connected

with a cable or wire to the headset. In some embodiments, a processor and a server are connected to the computer **910**. In some embodiments, an interactive keyboard is connected to the computer **910**. In some embodiments, an AI visual assistant **915** is active on the computer **910**. In some embodiments, a visual working agenda **960** is shown on the computer **910**. In some embodiments, user **905** can initiate and complete the business process with the visual assistant **905** via the computer **915** by the methods described in FIG. 1-FIG. 3. In some embodiments, an interactive panel is coupled to a central processor. In some embodiments, an interactive panel is coupled to a server via a wireless link. In some embodiments, the user **905** can choose what language to use. In some embodiments, other users can use this service described in this paragraph. In some embodiments, other users can use this service described in this paragraph. In some embodiments, the user is able to interact with multiple AI visual assistants as described in this example and methods described in FIG. 1-3.

[0065] FIG. 10 is a diagram showing a seventh example of a method for providing embodiments of the present disclosure.

[0066] In some embodiments, a user **1005** has a brain-computer interface. In some embodiments, the user **1005** may wear a headset **1007** that can detect and translate the electric signal from the brain and communicate with the computer or other devices. The computer **1010** or other devices are connected with wireless means to the headset. In some embodiments, a processor and a server are connected to the computer **1010**. In some embodiments, an interactive keyboard is connected to the computer **1010**. In some embodiments, an AI visual assistant **1015** is active on the computer **1010**. In some embodiments, a visual working agenda **1060** is shown on the computer **1010**. In some embodiments, user **1005** can initiate and complete the business process with the visual assistant **1005** via the computer **1015** by the methods described in FIG. 1-FIG. 3. In some embodiments, an interactive panel is coupled to a central processor. In some embodiments, an interactive panel is coupled to a server via a wireless link. In some embodiments, the user **1005** can choose what language to use. In some embodiments, other users can use this service described in this paragraph. In some embodiments, other users can use this service described in this paragraph. In some embodiments, the user is able to interact with multiple AI visual assistants as described in this example and methods described in FIG. 1-3.

Claims

1. A method for providing services via a set of virtual agents with artificial intelligence in a service area, the method comprising: detecting, by one or more processors, a parking request from a user, wherein the user may drive a vehicle into a garage of a mall; activating a first visual agent of the set of visual agents with artificial intelligence, wherein the first visual agent is configured to be a garage security personnel, wherein the first visual agent is configured to greet the user, give a parking spot and record location of the parking spot, wherein the first visual agent is configured to give directions of a first place that the user wants to visit, wherein the set of visual agents are empowered by an artificial intelligence engine, wherein the set of visual agents are configured to collaborate with each other, wherein an artificial intelligence engine is coupled to the one or more processors and a server, wherein the artificial intelligence engine is trained by human experts in the field, wherein the set of virtual agents are configured to be displayed in LED/OLED displays, Android/iOS tablets, Laptops/PCs, smartphones, or VWAR goggles, wherein a set of multi-layer info panels coupled to the one or more processors are configured to overlay graphics on top of the set of virtual agents, wherein any of the set of virtual agents are configured to be displayed with an appearance of a real human or a humanoid or a cartoon character, wherein any of the set of virtual agents' gender, age and ethnicity is determined by the artificial Intelligence's analysis on input from the user, wherein any of the set of virtual agents is configured to be displayed in full body or half body portrait mode, wherein the artificial intelligence engine is configured for real-time speech

recognition, speech to text generation, real-time dialog generation, text to speech generation, voice-driven animation, and human avatar generation, wherein the artificial intelligence engine is configured to emulate different voices and use different languages; activating, at the first place, by one or more processors, a second visual agent of the set of visual agents, when the user arrives via the directions of the first place given by the first visual agent, wherein the second visual agent is configured to be communicating with the first visual agent on info of the user, wherein the first place is a movie theater, wherein the second visual agent is configured to greet and display the movie time slots and seat availability in the theater, wherein the second visual agent is configured to recommend movies based on the users' past choices, wherein the second visual agent is configured to pick a time and a seat for the user for the movies based on user's choice, wherein the second visual agent is configured to give directions to a second place that the user wants to visit, wherein the second visual agent is configured to offer validation of parking ticket; activating, at the second place, by one or more processors, a third visual agent of the set of visual agents, when the user arrives via the directions of the second place given by the second visual agent, wherein the third visual agent is configured to be communicating with the first visual agent and the second visual agent on info of the user, wherein the second place is a restaurant, wherein the third visual agent is configured to greet and recommend movies based on the users' past choices, wherein the third visual agent is configured to complete the food transaction with the user, wherein the third visual agent is configured to give directions of destination chosen by the user; and activating the first visual agent of the set of visual agents with artificial intelligence, at the garage of the mall, when the user arrives at the garage, wherein the first visual agent is configured to recognize the user and the validated parking ticket, wherein the first visual agent is configured to say goodbye and welcome to visit us again in the future to the user when the user drives away.

2. A method for providing services via a set of virtual agents with artificial intelligence in a service area, the method comprising: activating a set of visual agents at different locations with artificial intelligence sequentially, wherein any of the visual agents are configured to greet the user, wherein the set of visual agents are empowered by an artificial intelligence engine, wherein the set of visual agents are configured to collaborate with each other, wherein an artificial intelligence engine is coupled to the one or more processors and a server, wherein the artificial intelligence engine is trained by human experts in the field, wherein the set of virtual agents are configured to be displayed in LED/OLED displays, Android/iOS tablets, Laptops/PCs, smartphones, or VR/AR goggles, wherein a set of multi-layer info panels coupled to the one or more processors are configured to overlay graphics on top of the set of virtual agents, wherein any of the set of virtual agents are configured to be displayed with an appearance of a real human or a humanoid or a cartoon character, wherein any of the set of virtual agents' gender, age and ethnicity is determined by the artificial Intelligence's analysis on input from the user, wherein any of the set of virtual agents is configured to be displayed in full body or half body portrait mode, wherein the artificial intelligence engine is configured for real-time speech recognition, speech to text generation, real-time dialog generation, text to speech generation, voice-driven animation, and human avatar generation, wherein the artificial intelligence engine is configured to emulate different voices and use different languages, wherein the set of visual agents are configured to collaborate with each other; enabling the set of visual agents at different locations with artificial intelligence sequentially to assist the user with a set of different requests, wherein the individual agent of the set of visual agents is configured to be optimized with tasks related to business that the individual agent is located; enabling the set of visual agents at different locations with artificial intelligence to exchange and analyze the information of the user to better assist the user; recording interactions between the user and the set of visual agents; and analyzing the interactions between the user and the set of visual agents to better assist the user in next encounter with the user.

3. A system for providing services via a set of virtual agents with artificial intelligence in a service area, the system comprising: a set of visual agents at different locations with artificial intelligence

sequentially, wherein any of the visual agents are configured to greet the user, wherein the set of visual agents are configured to collaborate among the set of visual agents, wherein the different locations are physical locations or digital locations, wherein the set visual agents are configured to collaborate with each other; an artificial intelligence engine that is coupled to the one or more processors and a server and the set of visual agents, wherein the set of visual agents are empowered by the artificial intelligence engine, wherein the artificial intelligence engine is trained by human experts in the field, wherein the set of virtual agents are configured to be displayed in LED/OLED displays, Android/iOS tablets, Laptops/PCs, smartphones, or VR/AR goggles, wherein a set of multi-layer info panels coupled to the one or more processors are configured to overlay graphics on top of the set of virtual agents, wherein any of the set of virtual agents are configured to be displayed with an appearance of a real human or a humanoid or a cartoon character, wherein any of the set of virtual agents' gender, age and ethnicity is determined by the artificial Intelligence's analysis on input from the user, wherein any of the set of virtual agents is configured to be displayed in full body or half body portrait mode, wherein the artificial intelligence engine is configured for real-time speech recognition, speech to text generation, real-time dialog generation, text to speech generation, voice-driven animation, and human avatar generation, wherein the artificial intelligence engine is configured to emulate different voices and use different languages, wherein the artificial intelligence engine is configured to perform following tasks: enabling the set of visual agents at different locations with artificial intelligence sequentially to assist the user with a set of different requests, wherein the individual agent of the set of visual agents is configured to be optimized with tasks related to business that the individual agent is located; enabling the set of visual agents at different locations with artificial intelligence to exchange and analyze the information of the user to better assist the user; recording interactions between the user and the set of visual agents; and analyzing the interactions between the user and the set of visual agents to better assist the user in next encounter with the user.
